

# CUET PG (SCQP09) PRACTICE QUESTION BANK

Computer Science & Information Technology | Subject: Data Structures

TOPIC: SPARSE MATRIX (MEDIUM  
LEVEL)

50 MCQs

**Instructions:**

- Each question carries 4 marks. For each incorrect response, 1 mark will be deducted (-1).
- Select the single best answer choice for each multiple-choice question.
- Answer Key is provided on the final page of this question sheet.

**Q1. A sparse matrix is strictly defined as a matrix in which:**

- (A) Number of zero elements > Number of non-zero elements      (B) Majority of elements are non-zero  
(C) Total non-zero elements  $O(N)$  where  $N$  is matrix order      (D) All diagonal elements are non-zero

**Q2. In a standard 3-column Triplet representation of an  $m \times n$  sparse matrix with  $k$  non-zero elements, the top header row (row index 0) stores:**

- (A)  $[0, 0, k]$       (B)  $[m, n, k]$   
(C)  $[m, n, m \times n]$       (D)  $[k, m, n]$

**Q3. If an  $8 \times 8$  matrix contains 10 non-zero elements, what is the space saving ratio when stored as 3-tuple array compared to a standard 2D array? (Assume integer takes 4 bytes)**

- (A) 33.3%      (B) 51.5%  
(C) 48.4%      (D) 65.6%

**Q4. What is the main structural disadvantage of storing a sparse matrix using standard 2D static arrays?**

- (A) High time complexity for element lookup      (B) Wastage of spatial memory due to zero storage  
(C) Inability to perform row traversal      (D) Incompatibility with recursive algorithms

**Q5. In Compressed Sparse Row (CSR) format, if a matrix has  $m$  rows and  $k$  non-zero elements, what is the size of the row\_ptr array?**

- (A)  $m$       (B)  $m + 1$   
(C)  $k$       (D)  $k + 1$

**Q6. In CSR representation, how is the number of non-zero elements in row  $i$  computed?**

- (A)  $\text{row\_ptr}[i]$       (B)  $\text{row\_ptr}[i+1] - \text{row\_ptr}[i]$   
(C)  $\text{row\_ptr}[i] + \text{row\_ptr}[i+1]$       (D)  $\text{col\_ind}[i+1] - \text{col\_ind}[i]$

**Q7. What is the asymptotic time complexity to transpose a sparse matrix represented in 3-tuple form using Fast Transpose algorithm?**

- (A)  $O(\text{cols} \times \text{non\_zero})$       (B)  $O(\text{rows} \times \text{cols})$   
(C)  $O(\text{cols} + \text{non\_zero})$       (D)  $O(\text{non\_zero}^2)$

**Q8. In Simple Transpose of a 3-tuple sparse matrix, what is the time complexity if matrix has  $C$  columns and  $K$  non-zero elements?**

- (A)  $O(C + K)$       (B)  $O(C \times K)$   
(C)  $O(C^2)$       (D)  $O(K \log K)$

**Q9. Which data structure extra arrays are required in Fast Transpose of a sparse matrix?**

- (A) row\_terms and start\_pos      (B) col\_terms and end\_pos  
(C) head\_node and next\_node      (D) frequency\_map and index\_stack

**Q10. In Fast Transpose, if row\_terms array stores count of elements in each column, what does start\_pos[i] represent?**

- (A) Number of elements before column  $i$       (B) Starting index of column  $i$  elements in transposed matrix  
(C) Total non-zero elements in column  $i$       (D) Both (A) and (B)

**Q11. Which linked list representation is best suited for dynamic operations (insert/delete non-zero entries) on a Sparse Matrix?**

- (A) Singly Linked List
- (B) Orthogonal Linked List (Cross-Linked List)
- (C) Circular Header List
- (D) Doubly Linked Stack

**Q12. In an Orthogonal Linked List representation of a Sparse Matrix, each node contains how many pointer fields?**

- (A) 1 pointer
- (B) 2 pointers (right, down)
- (C) 4 pointers (up, down, left, right)
- (D) 3 pointers

**Q13. Addition of two sparse matrices represented in 3-tuple form requires both matrices to have:**

- (A) Equal number of non-zero elements
- (B) Same dimensions ( $m \times n$ )
- (C) Same storage structure in CSR format
- (D) Square matrix order

**Q14. The time complexity of adding two sparse matrices A and B in 3-tuple format with non-zero elements  $K_1$  and  $K_2$  is:**

- (A)  $O(K_1 \times K_2)$
- (B)  $O(K_1 + K_2)$
- (C)  $O(m \times n)$
- (D)  $O(\max(K_1, K_2)^2)$

**Q15. In Compressed Sparse Column (CSC) format, which array corresponds directly to the row\_ptr array of CSR format?**

- (A) col\_ptr
- (B) row\_ind
- (C) val
- (D) dense\_vec

**Q16. Given CSR arrays for  $3 \times 3$  matrix:  $A=[5, 8, 3, 6]$ ,  $col\_ind=[0, 2, 1, 2]$ ,  $row\_ptr=[0, 2, 3, 4]$ . What is element at (0, 1)?**

- (A) 5
- (B) 8
- (C) 0
- (D) 3

**Q17. Referring to Q16, what is the entry in the matrix at position (1, 1)?**

- (A) 3
- (B) 8
- (C) 6
- (D) 0

**Q18. How many non-zero entries exist in row 0 of the matrix described in Q16?**

- (A) 1
- (B) 2
- (C) 3
- (D) 0

**Q19. Lower triangular matrix of size  $N \times N$  contains non-zero elements only where row index  $i$  and column index  $j$  satisfy:**

- (A)  $i < j$
- (B)  $i \geq j$
- (C)  $i = j$
- (D)  $i + j = N$

**Q20. Number of non-zero elements in an  $N \times N$  strictly upper triangular matrix (excluding main diagonal) is:**

- (A)  $N(N + 1) / 2$
- (B)  $N(N - 1) / 2$
- (C)  $N^2 / 2$
- (D)  $N - 1$

**Q21. For a symmetric matrix of order  $N \times N$ , minimum number of elements needed to store the matrix efficiently in 1D array is:**

- (A)  $N^2$
- (B)  $N(N + 1) / 2$
- (C)  $N(N - 1) / 2$
- (D)  $2N - 1$

**Q22. In row-major mapping of a Lower Triangular Matrix (1-based indexing), index in 1D array for element  $A[i][j]$  ( $i \geq j$ ) is:**

- (A)  $i(i - 1)/2 + j$
- (B)  $j(j - 1)/2 + i$
- (C)  $i \times N + j$
- (D)  $(i + j)/2$

- Q23. In column-major mapping of Lower Triangular Matrix (1-based indexing), index in 1D array for element  $A[i][j]$  ( $i \geq j$ ) is:**
- (A)  $[N(j-1) - (j-1)(j-2)/2] + (i-j+1)$  (B)  $i(i-1)/2 + j$   
 (C)  $N(i-1) + j$  (D)  $j(j+1)/2 + i$
- Q24. A tridiagonal matrix of size  $N \times N$  has non-zero elements only on diagonals where:**
- (A)  $|i-j| \leq 1$  (B)  $|i-j| > 1$   
 (C)  $i=j$  only (D)  $i+j = N$
- Q25. Total maximum number of non-zero elements in a Tridiagonal Matrix of order  $N \times N$  is:**
- (A)  $3N$  (B)  $3N-2$   
 (C)  $3N-1$  (D)  $N^2-3$
- Q26. A band matrix of size  $N \times N$  with bandwidth semi-width 'b' has non-zero elements for indices satisfying:**
- (A)  $|i-j| \leq b$  (B)  $|i-j| \geq b$   
 (C)  $i+j \leq b$  (D)  $i-j = b$
- Q27. Multiplication of a CSR sparse matrix ( $m \times n$  with  $K$  non-zeros) and a dense vector ( $n \times 1$ ) takes time:**
- (A)  $O(m \times n)$  (B)  $O(K)$   
 (C)  $O(m+K)$  (D)  $O(n \times K)$
- Q28. In 3-tuple array representation of Sparse Matrix, the array size needed for matrix with  $K$  non-zero elements is:**
- (A)  $K \times 3$  (B)  $(K+1) \times 3$   
 (C)  $m \times n$  (D)  $(K+1) \times 2$
- Q29. What is the space complexity of CSR representation for an  $m \times n$  matrix with  $K$  non-zero values?**
- (A)  $O(m+n+K)$  (B)  $O(m+K)$   
 (C)  $O(m \times n)$  (D)  $O(2K+m)$
- Q30. If a sparse matrix has zero elements at all places except the main diagonal, it is reduced to a:**
- (A) Scalar Matrix (B) Diagonal Matrix  
 (C) Identity Matrix (D) Tridiagonal Matrix
- Q31. How many elements are stored in a 1D array representation of a Diagonal Matrix of size  $N \times N$ ?**
- (A)  $N^2$  (B)  $N$   
 (C)  $2N-1$  (D)  $N(N-1)/2$
- Q32. In a 3-tuple sparse matrix, elements are traditionally sorted primarily by \_\_\_\_\_ and secondarily by \_\_\_\_\_.**
- (A) Row index, Column index (B) Column index, Row index  
 (C) Value, Row index (D) Value, Column index
- Q33. Which sparse storage format is most efficient for Column-based slicing and matrix operations?**
- (A) CSR (Compressed Sparse Row) (B) CSC (Compressed Sparse Column)  
 (C) DOK (Dictionary of Keys) (D) LIL (List of Lists)
- Q34. Dictionary of Keys (DOK) structure uses which core underlying data structure to map (row, col) to value?**
- (A) Binary Search Tree (B) Hash Map / Dictionary  
 (C) Priority Queue (D) Dynamic Array

**Q35. What is the time complexity of looking up an element at (r, c) in a DOK formatted sparse matrix?**

- (A)  $O(1)$  average
- (B)  $O(N)$
- (C)  $O(\log K)$
- (D)  $O(K)$

**Q36. List of Lists (LIL) representation of sparse matrix is constructed using:**

- (A) An array of lists containing (column, value) pairs per row
- (B) Matrix of pointers
- (C) Single circular queue
- (D) Double ended queue per column

**Q37. Which format is ideal for incrementally constructing a sparse matrix before converting to CSR/CSC?**

- (A) COO (Coordinate / Triplet format) or LIL
- (B) CSR
- (C) CSC
- (D) Dense Format

**Q38. In Cross-Linked List representation of a Sparse Matrix, header nodes exist for:**

- (A) Each row only
- (B) Each column only
- (C) Each row and each column
- (D) Only non-zero elements

**Q39. What is the total number of nodes (including header nodes) for an  $m \times n$  matrix with K non-zero elements in Cross-linked list?**

- (A)  $m + n + K$
- (B)  $m + n + K + 1$
- (C)  $m \times n$
- (D)  $K + 1$

**Q40. What is the main drawback of Fast Transpose algorithm over Simple Transpose algorithm?**

- (A) Greater time complexity
- (B) Requires auxiliary memory arrays
- (C) Destroys original matrix
- (D) Fails for non-square matrices

**Q41. In 0-based indexing for a  $5 \times 5$  Upper Triangular Matrix, row-major 1D array mapping index for  $U[i][j]$  ( $i \leq j$ ) is:**

- (A)  $i \times 5 - i(i - 1)/2 + (j - i)$
- (B)  $j(j + 1)/2 + i$
- (C)  $i(i + 1)/2 + j$
- (D)  $i \times 5 + j$

**Q42. The sparsity of a matrix is defined as:**

- (A) Number of non-zero elements / Total elements
- (B) Number of zero elements / Total elements
- (C) Total elements / Non-zero elements
- (D) Non-zero elements / Rows

**Q43. If a  $100 \times 100$  matrix has a density of 0.02, how many non-zero elements does it contain?**

- (A) 20
- (B) 200
- (C) 2000
- (D) 9800

**Q44. What is the density of a matrix if its sparsity is 0.85?**

- (A) 0.15
- (B) 0.85
- (C) 1.85
- (D) 0.30

**Q45. For a Sparse Toeplitz matrix of size  $N \times N$ , elements along each diagonal are constant. Unique elements stored are:**

- (A)  $N^2$
- (B)  $2N - 1$
- (C)  $N$
- (D)  $3N - 2$

**Q46. A square matrix A is symmetric if  $A = A^T$ . In sparse representation, storing upper triangle is sufficient because:**

- (A)  $A[i][j] = A[j][i]$
- (B) Lower triangle is all zero
- (C) Transpose equals inverse
- (D) Diagonals are all zero

**Q47. If row\_ptr array in CSR is [0, 3, 3, 5], how many non-zero entries are present in Row 1 (0-indexed)?**

- (A) 3 (B) 0  
(C) 2 (D) 5

**Q48. Referring to Q47, which row has the maximum number of non-zero elements?**

- (A) Row 0 (B) Row 1  
(C) Row 2 (D) Row 0 and Row 2 have equal maximum

**Q49. In a sparse matrix, element lookup in 3-tuple format using binary search (when sorted) takes time:**

- (A)  $O(1)$  (B)  $O(\log K)$  where K is non-zero count  
(C)  $O(K)$  (D)  $O(m \times n)$

**Q50. Which sparse matrix representation requires NO row or column pointers and directly stores tuples (i, j, value)?**

- (A) COO (Coordinate Format) (B) CSR  
(C) CSC (D) BSR (Block Sparse Row)

### ANSWER KEY (Q1 - Q50)

| Q. | Ans | Q. | Ans | Q. | Ans | Q. | Ans | Q. | Ans |
|----|-----|----|-----|----|-----|----|-----|----|-----|
| 1  | A   | 11 | B   | 21 | B   | 31 | B   | 41 | A   |
| 2  | B   | 12 | B   | 22 | A   | 32 | A   | 42 | B   |
| 3  | B   | 13 | B   | 23 | A   | 33 | B   | 43 | B   |
| 4  | B   | 14 | B   | 24 | A   | 34 | B   | 44 | A   |
| 5  | B   | 15 | A   | 25 | B   | 35 | A   | 45 | B   |
| 6  | B   | 16 | C   | 26 | A   | 36 | A   | 46 | A   |
| 7  | C   | 17 | A   | 27 | C   | 37 | A   | 47 | B   |
| 8  | B   | 18 | B   | 28 | B   | 38 | C   | 48 | A   |
| 9  | A   | 19 | B   | 29 | B   | 39 | B   | 49 | B   |
| 10 | D   | 20 | B   | 30 | B   | 40 | B   | 50 | A   |